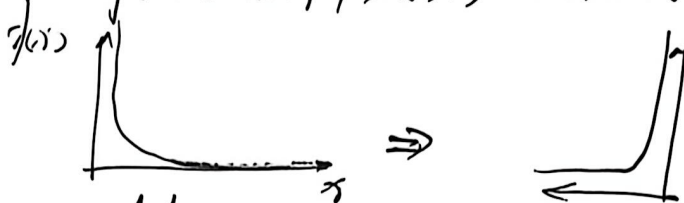


lecture 数据分布有“长尾” (即中位数远超过平均值) 时, 其期望值可能非常大.

e.g. $\Pr(D \geq r) = 1/r$

$$\sum_{n=1}^{\infty} \frac{1}{n} \Rightarrow \int_1^{\infty} \frac{1}{x} dx = \ln(x) \Big|_1^{\infty} = \infty$$

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Plugging (18.6) and (18.7) into (18.5):

$$\begin{aligned} \text{Ex}[C] &= 1 \cdot p + (1 + \text{Ex}[C])(1 - p) \\ &= p + 1 - p + (1 - p) \text{Ex}[C] \\ &= 1 + (1 - p) \text{Ex}[C]. \end{aligned}$$

Can that problem

n个人的帽子打乱, 拿回自己帽子的期望

看初始变量 X_i : $\begin{cases} 1, & \text{if } i=1 \\ 0, & \text{if } i \neq 1 \end{cases}$

Then, rearranging terms gives

$$1 = \text{Ex}[C] - (1 - p) \text{Ex}[C] = p \text{Ex}[C],$$

and thus

$$\boxed{\text{Ex}[C] = 1/p.}$$

→ 每一步独立概率为 p 的情况下

The general principle here is well-worth remembering.

Mean Time to Failure

If a system independently fails at each time step with probability p , then the expected number of steps up to the first failure is $1/p$.

So, for example, if there is a 1% chance that the program crashes at the end of each hour, then the expected time until the program crashes is $1/0.01 = 100$ hours.

As a further example, suppose a couple insists on having children until they get a boy, then how many baby girls should they expect before their first boy? Assume for simplicity that there is a 50% chance that a child will be a boy and that the genders of siblings are mutually independent.

This is really a variant of the previous problem. The question, "How many hours until the program crashes?" is mathematically the same as the question, "How many children must the couple have until they get a boy?" In this case, a crash corresponds to having a boy, so we should set $p = 1/2$. By the preceding analysis, the couple should expect a baby boy after having $1/p = 2$ children. Since the last of these will be a boy, they should expect just one girl. So even in societies where couples pursue this commitment to boys, the expected population will divide evenly between boys and girls.

There is a simple intuitive argument that explains the mean time to failure formula (18.8). Suppose the system is restarted after each failure. This makes the mean time to failure the same as the mean time between successive repeated failures. Now if the probability of failure at a given step is p , then after n steps we expect to have pn failures. Now, by definition, the average number of steps between failures is equal to np/p , namely, $1/p$.

转来问题: 即皮百佳
构也亦然

线性性 ~~→~~
↑